

Anomaly Detection : Phase 2

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Dataset Selection

For this project we have selected the Kvasir Dataset (V3). This dataset contains endoscopy images taken from the gastrointestinal tract during medical examinations. The dataset includes both normal (healthy) images and images with different diseases, which makes it suitable for an anomaly detection project.

Anomaly detection using autoencoders works by training the model on normal images only so that it learns what healthy tissue looks like. When the model later sees abnormal images, it usually reconstructs them poorly, which produces a higher reconstruction error. These differences can help identify abnormal regions in the images.

All images used in this project were resized to 512×512 pixels, which is the required input size for the DeepTune autoencoder model.

Healthy / Normal Images

Healthy images represent normal gastrointestinal tissue where no visible disease is present. These images will be used to train the autoencoder so the model can learn the normal appearance of the tissue.

The healthy dataset was divided into three groups:

- Healthy – Train: 1398 images
- Healthy – Baseline: 199 images
- Healthy – Test: 401 images

The training set is used to train the autoencoder. The baseline set will later be used to compute the average reconstruction error for healthy images. The test set will be used to compare normal images against pathological images.

The total number of healthy images used in this project is 1998 images.

Pathological Images

The dataset also contains images showing different gastrointestinal diseases. These pathological images will be used to evaluate how well the anomaly detection model can identify abnormal tissue.

The pathological conditions included in this project are:

- Esophagitis – inflammation of the esophagus
- Polyps – abnormal tissue growths that can develop into cancer
- Ulcerative Colitis – a chronic inflammatory disease affecting the colon

Number of images for each pathology:

- Esophagitis: 1000 images
- Polyps: 994 images
- Ulcerative Colitis: 990 images

The total number of pathological images used is 2984 images.

Project Plan

In this project, an autoencoder model will be trained using only the healthy training images. After training, the model will attempt to reconstruct images from the healthy test dataset and from the pathological datasets.

The difference between the original image and the reconstructed image will be measured as the reconstruction error. Areas with higher reconstruction error are likely to represent structures that the model did not learn from healthy images. These areas may correspond to abnormalities or diseases in the images.

This approach allows the model to highlight potential anomalies in medical images without needing labeled training data for each disease.

Methods

This stage involved creating an anomaly detection framework with DeepTune's autoencoder. The concept behind autoencoders is training the model on only healthy images so that it learns the features of normal tissue. Resizing was performed using DeepTune preprocessing functions. When it encounters abnormal images, the model fails to reconstruct the abnormal images accurately leading to a higher reconstruction error. We use this error metric to detect anomalies.

Preparing and resizing data:

We converted the Kvasir dataset to parquet format which was easier for us to read and manipulate. We then created 4 different parquets corresponding to each group of images. One

each for healthy images and the other three for each pathology i.e. esophagitis, polyps and ulcerative colitis.

We then resized all the images to 512×512 . This was done using the newly improved DeepTune function for resizing images. Higher image sizes were chosen to reduce loss of information during resizing.

Splitting datasets:

We split only the healthy dataset into 3 different datasets.

- Healthy-Train (70%)
- Healthy- baseline (10%)
- Healthy-Test (20%)

The train dataset is what we would use to train our autoencoder. The baseline dataset would be used to calculate the mean error of normal/reconstruction images later on. The test dataset would be used to hold our normal dataset for comparisons with the pathological datasets.

Autoencoder Training

We only trained the autoencoder on the healthy train dataset. We opted to use the deeper autoencoder model provided by DeepTune to help with learning.

Training Parameters:

- Epochs: 50
- Learning rate: 0.0001
- Image size: 512×512
- Deeper Model: True (--use-deeper)

During autoencoder training, the model attempts to learn how to reduce the input image into a smaller representation which it then learns to reconstruct back into the original image.

Image Reconstruction

Post training, we passed in images from healthy-baseline, healthy-test and all pathological images to be reconstructed by the model.

We then saved each original image and it's reconstructed version as .png images in the required <P#>-<V#>-original.png and <P#>-<V#>-reconstructed.png format.

Reconstruction Error Calculation

To calculate the reconstruction error we found the absolute difference between the original image and it's reconstruction per pixel.

This should highlight regions in the image where there is high probability of abnormality as the model fails to reconstruct those areas. We saved these images as <P#>-<V#>-reconstruction_error.png

Baseline Error Computation

To make the detection more reliable, a baseline error image was created using the healthy-baseline dataset.

All reconstruction error images from this dataset were averaged pixel by pixel to create a single image:

- average_reconstruction_error_healthy_baseline.png

This image represents the normal level of reconstruction error expected for healthy images.

Anomaly Image Generation

For each image in the healthy-test and pathological datasets, a final anomaly image was created by subtracting the baseline error from the reconstruction error.

In simple terms, this shows how much an image differs from normal patterns. Larger differences indicate potential anomalies.

These images were saved as:

- <P#>-<V#>-subtraction_reconstruction_error.png

Scaling of Anomaly Images

Since the raw anomaly images can be hard to compare, a global scaling method was used.

All images were scaled using the same maximum value (**246.0**) so that the brightness levels are consistent across all images. This makes it easier to visually compare healthy and pathological cases.

The scaled images were saved as:

- <P#>-<V#>-scaled_subtraction_reconstruction_error.png